

MinJun Chang

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Education

Georgia Institute of Technology, Ph.D. in Civil Engineering	Aug 2025 - Current
• RICAL Group (Dr. Yong Cho) & HARMONIC Lab (Dr. Francis Baek)	
Yonsei University, B.S. in Mechanical Engineering	Mar 2019 - Feb 2025

Research

RICAL Group, Georgia Institute of Technology – Atlanta, GA, USA	Aug 2025 – Current
Graduate Research Assistant, Supervisor: Dr. Yong K. Cho	
• Developing IMU-based worker motion recognition and zone-level localization for construction safety	
• Designing perception-to-decision systems for vehicle monitoring and risk assessment in work zones	
HARMONIC Group, Georgia Institute of Technology – Atlanta, GA, USA	Aug 2025 – Current
Graduate Research Assistant, Supervisor: Dr. Francis Baek	
• Developing humanoid motion control for dexterous tasks in construction and manufacturing environments	
• Studying human-robot interaction and practitioner perception of robotic automation for field deployment	
Dynamic Robotic System Laboratory, Seoul National University – Seoul, KR	Jul 2024 – Dec 2024
Undergraduate Research Assistant, Supervisor: Dr. Jaeheung Park	
• Model-free reinforcement learning framework with a state estimator neural-network for bipedal locomotion	

Publications

1. **Chang, M.**, Kim, T., Cho, Y., Baek, F. (2026). “Beyond Human-Derived Motion: Exploring Robot-Specific Execution for Construction Humanoids” *IROS*, Workshop on the Future of Construction, Under Review.
2. Kim, T., Lee, G., **Chang, M.**, Lim, S., Choi, J. (2026). “Deployable Human Preference Alignment in Robotics: Learning Representative Rewards from Diverse Human Preferences.” *AAAI 2027*, Submitted, *arXiv:2607.12466*.
3. **Chang, M.**, Machiraju, S., Baek, F., Cho, Y. (2026). “Real-Time Temporal Convolution based Worker Motion Recognition with Three IMU Sensor Combinations.” *ISARC 2026*, Singapore.
4. **Chang, M.**, Machiraju, S., Cho, Y. (2026). “Field Evaluation of a Wearable Framework for Worker Motion Recognition and Zone-Level Localization.” *ISARC 2026*, Singapore.
5. Baek, F., **Chang, M.**, Kim, E. (2026). “Large Language Models as Virtual Personas to Estimate Construction Practitioners’ Perceptions of Robotic Automation.” *ASCE i3CE 2026*, Songdo, Korea.
6. Kim, S., **Chang, M.**, Machiraju, S., Cho, Y. (2026). “A Perception-to-Decision Vehicle Monitoring System for Highway Work-Zone Safety.” *ASCE i3CE 2026*, Songdo, Korea.
7. **Chang, M.**, Shin, J., Park, J. (2025). “Robust Symmetric Bipedal Locomotion Development via Simultaneous State Estimator Neural Network Training.” *KRoC 2025*, Jeju, Korea.
8. Nam, H.*, Choi, S.*, **Chang, M.***, Yang, J.*, Lim, J., Choi, J. (2023). “State Prediction based Control Input Delay Compensation for Autonomous Driving Systems.” *KRoC 2023*, Jeju, Korea. *equal contribution.

Patents and Copyrights

The Urine Examination Apparatus and Controlling Method of the Same	Mar 2026
KR10-2944428-00-00, Registered-Active	
Autonomous Driving Auxillary Cart Robot for Manufacture	Aug 2024
KR10-2024-0177135, under prosecution	
Eye Tuner: Media Literacy Program based on Pupil Tracking by Computer Vision	Nov 2021
Korea Copyright Commission, C-2024-039138, Registered-Active	

Professional Experience

Hanwha Systems(Space&Defense) , Satellite Software Engineer	Jan 2025 – May 2025
• ARMv7 MCU internal communication software programming for Small Synthetic Aperture Radar (SAR) Satellite	
GOLE Robotics , Robotics Engineer	Apr 2024 – Jun 2024
• Implemented global and local robot path planning algorithm on ROS2 for construction delivery robot WERO • Developed actuator controller package with C++/Python binding enabling python API usage of C++ source	
DRIMAES , Embedded Software Engineer	Oct 2022 – Mar 2024
• Linux, ARM MCU software/firmware programming for various products • Developed various communication protocols (Serial, MQTT, REST, CAN) • Implemented multiple virtual container management technique on cross-platform systems	
SIOT Infotech , Embedded Software Engineer	Feb 2022 – Oct 2024
• Developed embedded software for Mediatek, ESP chipsets and enhanced custom OpenWRT OS kernel	

Selected Awards

1st place, National ICT Smart Device Competition , Korean Ministry of Science and ICT	Aug 2024
• Awarded by the Minister of Science and ICT of Republic of Korea • Led a team of 5 in developing an Autonomous Manufacture Assistant CARTRASCHE	
1st place, 2022 Autonomous Driving Robot Racing Contest , Korean Robotics Society (KRoS)	Nov 2022
• Participated as Localization team member • Developed a control algorithm utilizing LiDAR, IMU, and GPS sensor fusion for collision avoidance	
Selection, Hanium Contest , Federation of Korea Information Industries	Nov 2021
• Led a team of 4 in developing Personalized Content Literacy program EYE-TUNER • Implemented pupil tracking algorithm for the program	

Projects

Awaresite : Intelligent PPU System for Construction Productivity and Safety	Aug 2025 - Current
• Deep-Learning based worker motion recognition using IMU data • Zone scale indoor localization using BLE beacons	
CARTRASCHE: Autonomous Driving Auxiliary Cart Robot	Mar 2023 - Jul 2024
• Developed autonomous driving mobile robot with rotating shelf system using SLAM for navigation in ROS • Implemented custom RC filter and encoder-less motor control algorithm for activation • Managed the project flow and system overall management • Tools Used: C++, Python, ROS, LinuxOS	
UAV Fleet Management System for Robot Cluster in Factory	Aug 2023 - Dec 2023
• Implemented task scheduling and allocation algorithm based on order status for multi-robot network • Tools Used: Python, ROS, MQTT	
FennecBot: Industrial Anomaly Detection Mobile Robot	May 2023 - Aug 2023
• Developed multi-modal deep learning network for pipeline anomaly detection and the classification of pipeline leakage using RGB camera, and ultrasonic/acoustic sound camera • Operated on Scout mini with line-tracing algorithm detecting pipe leakage within Hyundai HI. factory	

Honors

Federico Stubbe Graduate Student Fellowship , Georgia Institute of Technology	Aug 2025
Fellowship granted to competent graduate students	

Honor society for outstanding young engineers in Republic of Korea

Technologies

Programming: Python, C++, C, JavaScript, MATLAB

Frameworks/Tools: ROS, IsaacSim/Lab, PyTorch, Docker, AWS, FastAPI, Solidworks, Fusion360

Hardware: ARM V7, Jetson Xavier, Jetson Nano, RaspberryPi, Arduino, STM32, ESP32, Unitree Go1

Languages: Korean(Native), English(Fluent), Japanese(Median)